Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy by dataset.

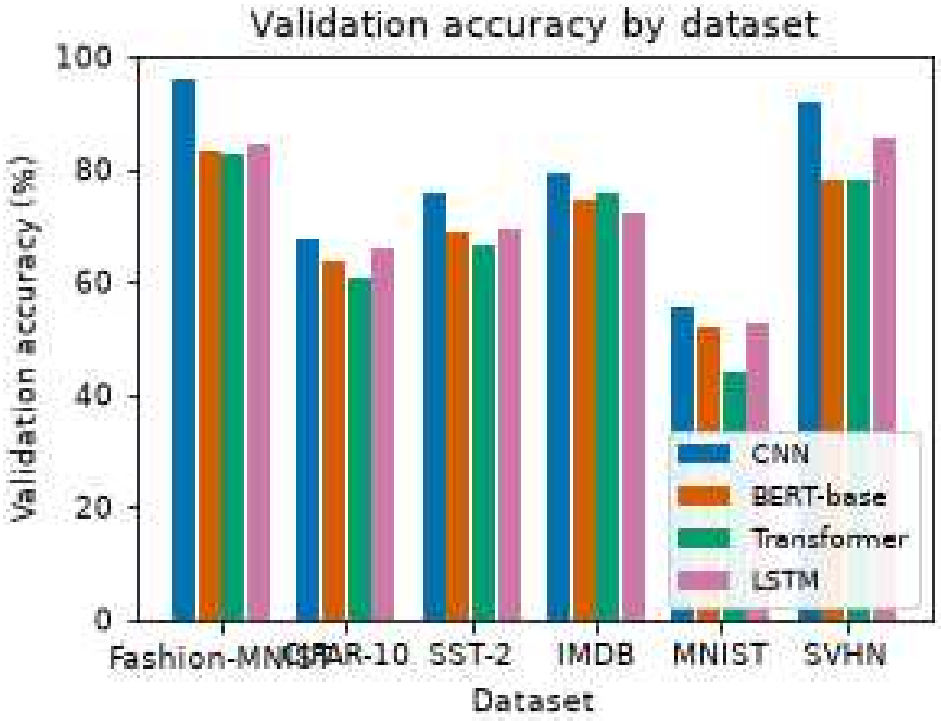


Figure 1: Validation accuracy by dataset for the models in this experiment.

LSTM reaches 69.3% validation accuracy on SST-2. LSTM achieves about 90% validation accuracy on SVHN (Figure 1).

Figure 2 shows the accuracy during training.

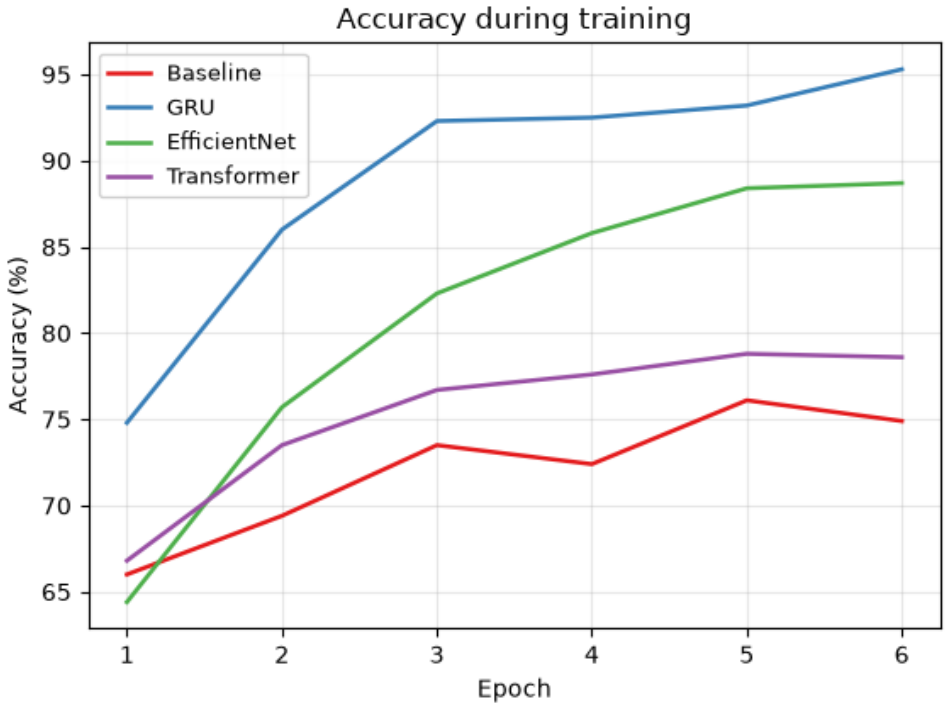


Figure 2: Accuracy during training for the models in this experiment.

As shown in Figure 2, at epoch 1, GRU achieves 68.0% accuracy. GRU outperforms Transformer by 14.2 percentage points at epoch 4.

Figure 3 shows the accuracy during training.

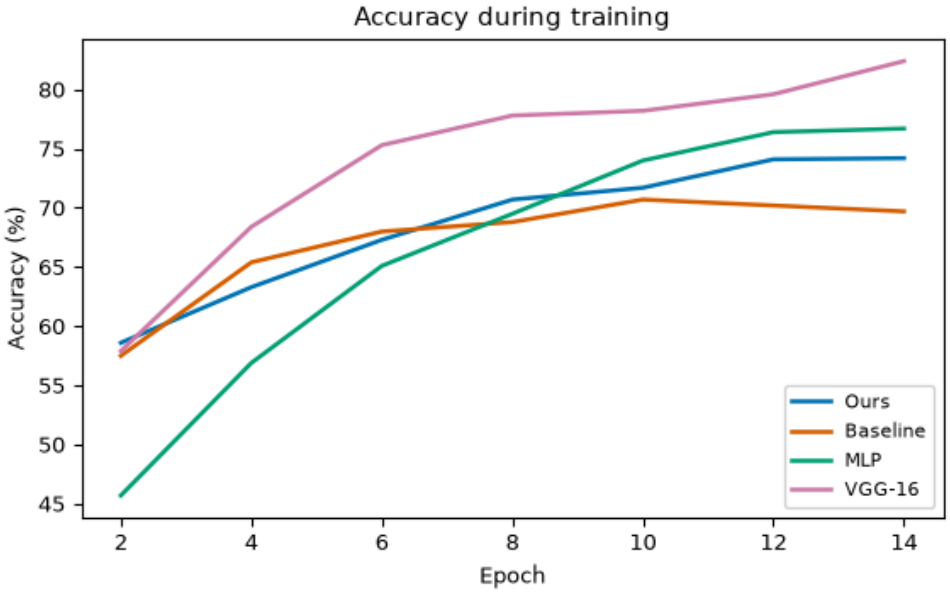


Figure 3: Accuracy during training for the models in this experiment.

At epoch 8, MLP achieves 69.5% accuracy. At epoch 14, Ours achieves 82.3% accuracy.

3 Discussion

We note that the training was stable for all models.